Hypothesis testing framework

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General framework for hypothesis tests

Definition: Let $\theta \in \mathbb{R}^p$ be some parameter of interest. A **hypothesis** is a statement about θ. The goal of a hypothesis test is to compare two competing hypotheses:

$$H_0 : \theta \in \Theta_0 \qquad H_A : \theta \in \Theta_1$$

- $\Theta_0 \cap \Theta_1 = \emptyset$ (disjoint) $\qquad \Theta_0 \cup \Theta_1 = \Theta$ ← full parameter space (exhaustive)

- If $\Theta_0 = \{\theta_0\}$, H_0 is a <u>simple</u> hypothesis
  Otherwise, H_0 is a <u>composite</u> hypothesis
  (likewise for H_A)

Ex: $H_0 : \begin{pmatrix} \beta_3 \\ \beta_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\qquad H_A : \begin{pmatrix} \beta_3 \\ \beta_4 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

simple null $\qquad$ composite alternative

Outcomes

$$H_0 : \theta \in \Theta_0 \qquad H_A : \theta \in \Theta_1$$

The test either **rejects** H_0 or **fails to reject** H_0. Possible outcomes:

Decision \ Truth	H_0 is true	H_0 is false
fail to reject	✓	type II error (false negative)
reject	type I error (false positive)	✓

Goal

	H_0 is true	H_0 is false
fail to reject	correct decision	type II error
reject	type I error	correct decision

Goal: Minimize type II error rate, subject to control of type I error rate

Constructing a test

$$H_0 : \theta \in \Theta_0 \qquad H_A : \theta \in \Theta_1$$

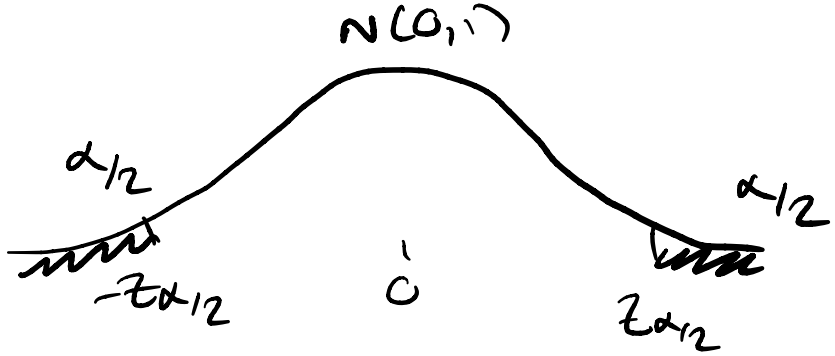
- choose α (desired type I error rate)
- observe data $X_1, \ldots, X_n$
- calculate test statistic $T_n = T(X_1, \ldots, X_n)$
- choose a rejection region (typically depends on α)

$$R = \{(x_1, \ldots, x_n) : \text{reject } H_0\}$$

(often write in terms of T_n)

- Reject H_0 if $(X_1, \ldots, X_n) \in R$
  (otherwise fail to reject)

Constructing a test



Given observed data $X_1, ..., X_n$:

1. Calculate a test statistic $T_n = T(X_1, ..., X_n)$
2. Choose a rejection region $\mathcal{R} = \{(x_1, ..., x_n) : \text{ reject } H_0\}$
3. Reject H_0 if $(X_1, ..., X_n) \in \mathcal{R}$

Example: $X_1, ..., X_n$ iid with mean μ and variance σ^2 (assume σ known)

$H_0: \mu = \mu_0$ $\quad$ $H_A: \mu \neq \mu_0$

① $T(X_1, \dots, X_n) = \dfrac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}$

② $\mathcal{R} = \left\{ (x_1, \dots, x_n) : \left| \dfrac{\bar{x}_n - \mu_0}{\sigma/\sqrt{n}} \right| > z_{\frac{\alpha}{2}} \right\}$

rejection region determined by desired type I error rate

Prob. of type I error $= P(\text{reject } H_0 \mid H_0 \text{ true})$

$$= P\left(\left| \underbrace{\frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}}}_{\approx N(0,1)} \right| > z_{\alpha/2} \;\middle|\; \mu = \mu_0 \right) \approx \alpha$$

Power function

Suppose we reject H_0 when $(X_1, ..., X_n) \in R$. The **power function** $\beta(\theta)$ is

$$\beta(\theta) = \underbrace{P_\theta((X_1, ..., X_n) \in R)}_{\text{probability of rejecting } H_0 \text{ if true parameter is } \theta}$$

function of θ

Goal: want $\beta(\theta)$ small when $\theta \in \Theta_0$
and $\beta(\theta)$ large when $\theta \in \Theta_1$

Formally:
1. Fix $\alpha \in [0,1]$
2. Try and maximize $\beta(\theta)$ for $\theta \in \Theta_1$ subject to $\beta(\theta) \leq \alpha$ for $\theta \in \Theta_0$

- If $\sup_{\theta \in \Theta_0} \beta(\theta) = \alpha$, our test is size α
- If $\sup_{\theta \in \Theta_0} \beta(\theta) \leq \alpha$, our test is level α

Example

$\sqrt{n}(\bar{X}-\mu) \approx N(0,\sigma^2) \Rightarrow \bar{X}_n \approx N(\mu, \frac{\sigma^2}{n})$

$\Rightarrow \frac{\bar{X}_n-\mu}{\sigma/\sqrt{n}} \approx N(0,1)$

$X_1, \ldots, X_n$ iid from a population with mean μ and variance σ^2.

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

$$\beta(\mu) = P_\mu\left(\left| \frac{\bar{X}_n-\mu_0}{\sigma/\sqrt{n}} \right| > z_{\frac{\alpha}{2}} \right)$$

$$= \underbrace{P_\mu\left(\frac{\bar{X}_n-\mu_0}{\sigma/\sqrt{n}} > z_{\frac{\alpha}{2}} \right)} + P_\mu\left(\frac{\bar{X}_n-\mu_0}{\sigma/\sqrt{n}} < -z_{\frac{\alpha}{2}} \right)$$

$$P_\mu\left(\frac{\bar{X}_n-\mu+\mu-\mu_0}{\sigma/\sqrt{n}} > z_{\alpha/2} \right)$$

$$\hookrightarrow P_\mu\left(\underbrace{\frac{\bar{X}_n-\mu}{\sigma/\sqrt{n}}}_{\approx N(0,1)} > z_{\alpha/2} - \left(\frac{\mu-\mu_0}{\sigma/\sqrt{n}}\right) \right)$$

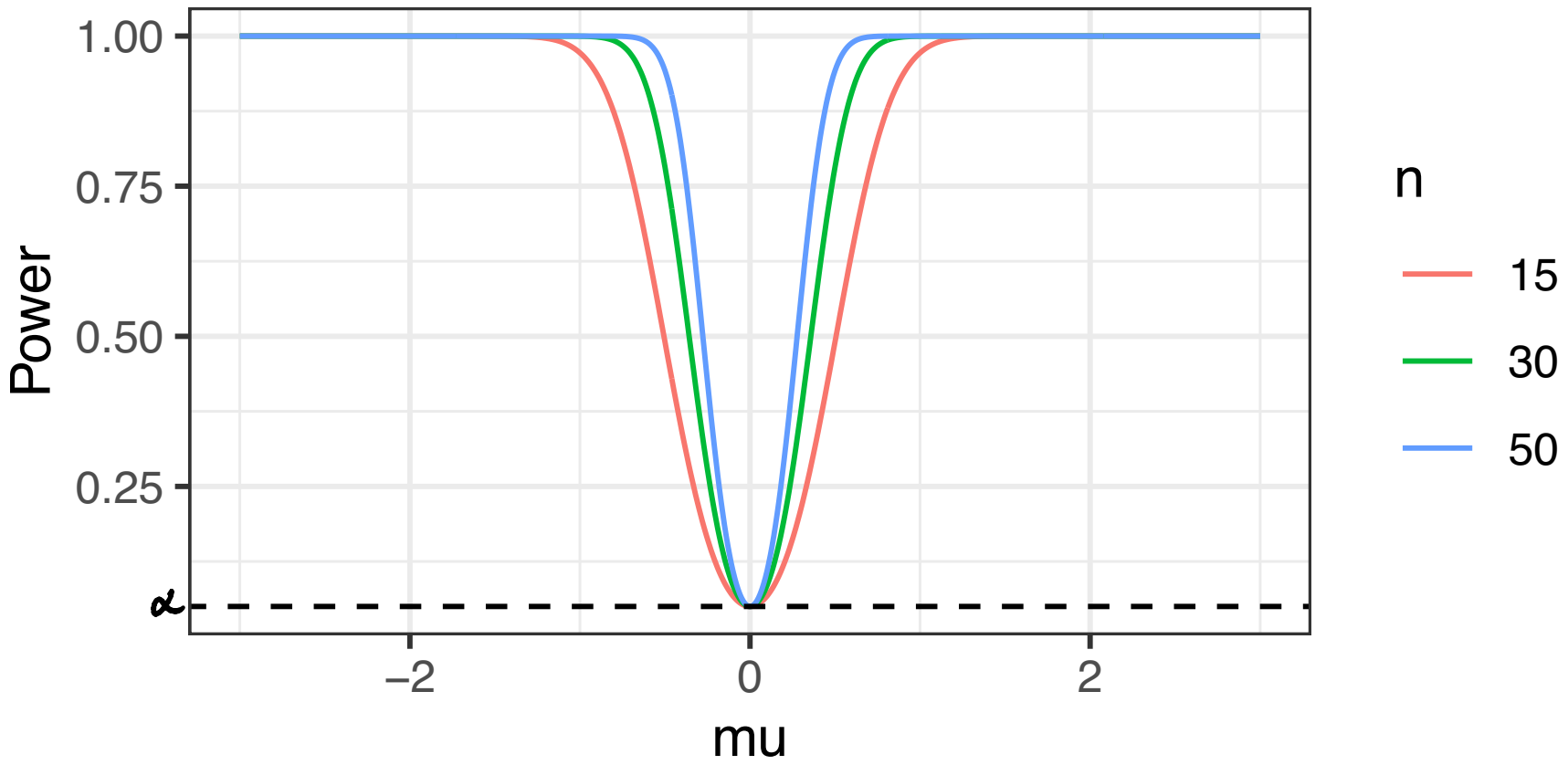
$$\Rightarrow \beta(\mu) \approx 1 - \underset{\substack{\uparrow \\ N(0,1)\text{ cdf}}}{\Phi}\left(z_{\alpha/2} - \left(\frac{\mu-\mu_0}{\sigma/\sqrt{n}}\right)\right) + \Phi\left(-z_{\alpha/2} - \left(\frac{\mu-\mu_0}{\sigma/\sqrt{n}}\right)\right)$$

Example

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

$$\beta(\mu) \approx 1 - \Phi\left(z_{\alpha/2} - \left(\frac{\mu - \mu_0}{\sigma/\sqrt{n}}\right)\right) + \Phi\left(-z_{\alpha/2} - \left(\frac{\mu - \mu_0}{\sigma/\sqrt{n}}\right)\right)$$

Example: $\sigma = 1$, $\mu_0 = 0$



Example

$X_1, ..., X_n$ iid from a population with mean μ and variance σ^2.

$$H_0 : \mu = \mu_0 \qquad H_A : \mu \neq \mu_0$$

$$\beta(\mu) \approx 1 - \Phi\left(z_{\alpha/2} - \left(\frac{\mu-\mu_0}{\sigma/\sqrt{n}}\right)\right) + \Phi\left(-z_{\alpha/2} - \left(\frac{\mu-\mu_0}{\sigma/\sqrt{n}}\right)\right)$$

For any n: if $\mu = \mu_0$ (H_0 is true)

$$\beta(\mu) \approx 1 - \Phi\left(z_{\frac{\alpha}{2}}\right) + \Phi\left(-z_{\alpha/2}\right) = \frac{\alpha}{2} + \frac{\alpha}{2} = \alpha$$

(type I error rate)

<u>Fix n</u>: as $|\mu - \mu_0| \to \infty$: $\beta(\mu) \to 1$

as $z_{\alpha/2} - \left(\frac{\mu-\mu_0}{\sigma/\sqrt{n}}\right) \to -\infty$,

$$\Phi\left(z_{\alpha/2} - \left(\frac{\mu-\mu_0}{\sigma/\sqrt{n}}\right)\right) \to 0$$

$$\beta(\mu) \to 1 - 0 + 0 = 1$$

<u>Fix $\mu \neq \mu_0$</u>: as $n \to \infty$, $\beta(\mu) \to 1$

Rejecting H_0

$$H_0 : \theta \in \Theta_0 \qquad H_A : \theta \in \Theta_1$$

Question: A hypothesis test rejects H_0 if $(X_1, ..., X_n)$ is in the rejection region $\mathcal{R}$. Are there any issues if we only use a rejection region to test hypotheses?

rejection region: only gives binary info
(in region or not)

p-values

$$H_0 : \theta \in \Theta_0 \qquad H_A : \theta \in \Theta_1$$

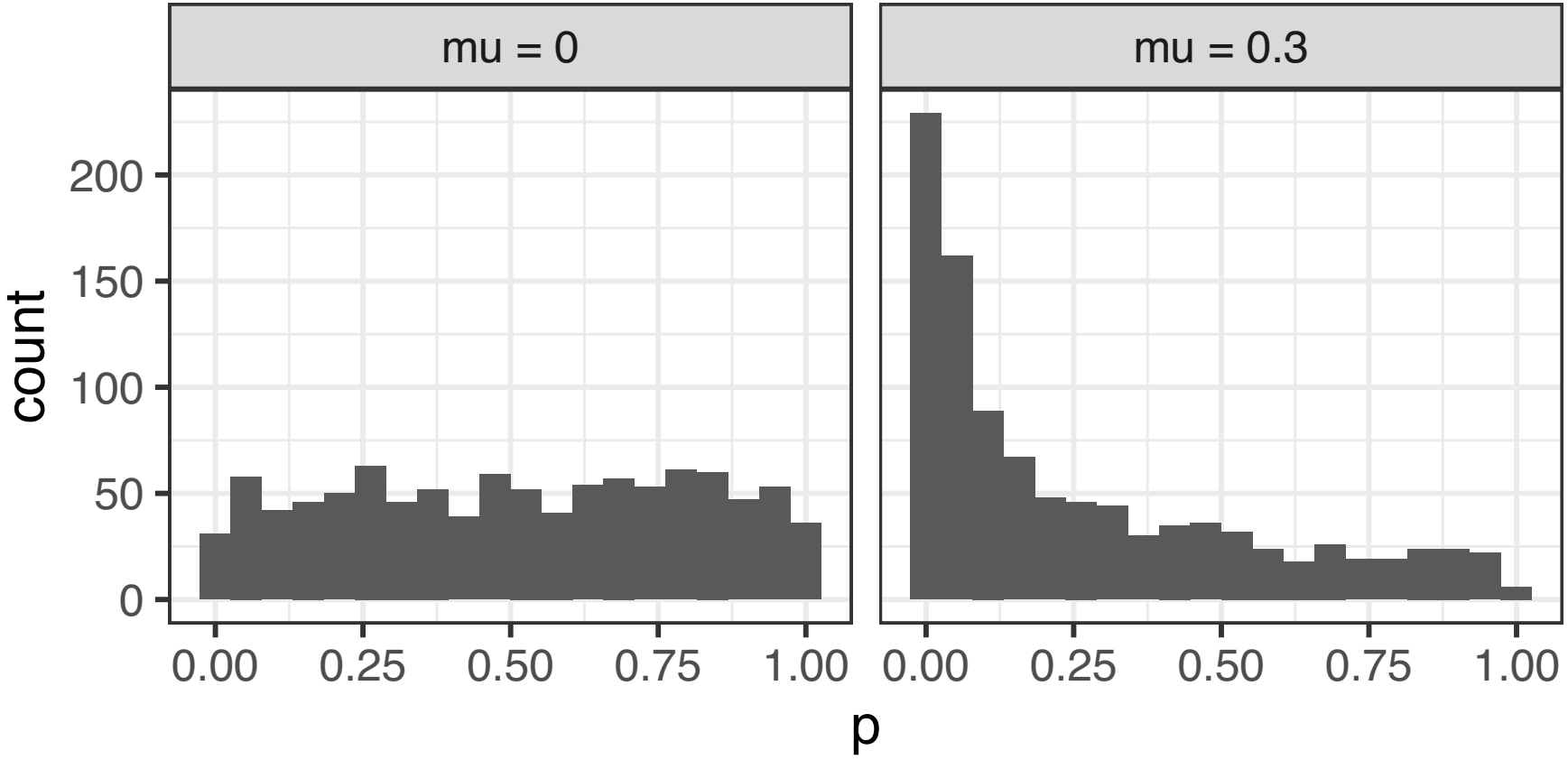
A **p-value** $p(X_1, ..., X_n)$ is a test statistic satisfying:

- $0 \leq p(X_1, \ldots, X_n) \leq 1$
- small values of $p(X_1, \ldots, X_n)$ are evidence against H_0
  (reject H_0 for sufficiently small p-values)
- For all $\theta \in \Theta_0$ and all $\alpha \in [0,1]$:
  $$P_\theta(p(X_1, \ldots, X_n) \leq \alpha) \leq \alpha$$
  (if we reject when $p \leq \alpha$, then $P(\text{type I error}) \leq \alpha$)

Example: p-values

Generate data $X_1, ..., X_{20} \overset{iid}{\sim} N(\mu, 1)$. Test $H_0 : \mu = 0$ vs. $H_A : \mu \neq 0$.

Distribution of p-values:



p-values

Suppose we have a test which rejects H_0 when $T(X_1, ..., X_n) > c_\alpha$, where c_α is chosen so that

$$\sup_{\theta \in \Theta_0} \beta(\theta) = \sup_{\theta \in \Theta_0} P_\theta(T(X_1, ..., X_n) > c_\alpha) \leq \alpha$$

Let $x_1, ..., x_n$ be a set of observed data.

Theorem: Let

$$p(x_1, ..., x_n) = \sup_{\theta \in \Theta_0} P_\theta(T(X_1, ..., X_n) > T(x_1, ..., x_n))$$

Then p is a valid p-value.